ECHO-7

Visual bay finding and docking | Product & operating guide

DE1-SoC / OV7670 / UGV02

Choose a blue bay marker and start a task. ECHO-7 searches, confirms the marker, aligns, approaches and parks, with live VGA feedback, remote stop and front sonar protection.

OV7670 camera — Live colour image → **DE1-SoC FPGA** — Cyclone V 5CSEMA5F31C6 | vision + control → **UGV02 chassis** — Differential-wheel motion

Also fitted: J3 RCWL-1601 sonar, NEC IR receiver/remote, VGA monitor, switches, keys, LEDs and six-digit display. USB-Blaster provides PC download/debug. Chassis UART: 115200 8N1, 3.3 V logic, common ground.

THREE OPERATING MODES

MODE / DISPLAY CODE	PURPOSE
0 - INTEGRATED	Camera-guided driving. Choose one bay; POWER or red2 starts a new task. Default after power-up/reset, with motion disarmed.
1 - VISION_ONLY	Live recognition, tracking, calibration and VGA; wheel commands stay zero. Use for marker and camera setup.
2 - CHASSIS_INJECT	PC/JTAG virtual targets test the same controller. Real IR and sonar stay active; loss of the PC injection session stops motion.

Choose modes with the JTAG debug tool. Changing mode stops the task. INTEGRATED needs no PC heartbeat.

RED AND BLUE MARKERS

Blue 1-6: destination bays. SW0-SW5 choose one number to follow. Keep its complete border and white anchor visible.

Red1: colour calibration. Hold red1 and its paired blue reference stripe steady. Motion pauses until calibration finishes: **CAL:Y**. The stripe label **BJ** is not a bay.

Red2: reselect and start. Re-read switches, relock and request a task. Remove it, then present it beside the target. It cannot choose a bay or restart an active task. Keep hands/cards clear of J3.

SWITCHES AND BUTTONS

SW0-SW5	SW0/1/2/3/4/5 select blue 1/2/3/4/5/6. **Exactly one ON** to start. Changing selection cancels motion.
SW6	Candidate colour/number diagnostics.
SW7	Centre colour sample and marker identity.
SW8	0: normal perspective handling. 1: bounding-box recognition bypass.
SW9	Camera-only display; cancels motion. OFF requires a new start to move again.
KEY0 / KEY1	KEY0: full reset. KEY1: next display page.

Remote POWER: start. **A:** latched emergency stop. After A, one fresh POWER releases it; release and press again to start. Holding POWER cannot do both.

PC/JTAG: In stopped, disarmed VISION_ONLY, use **tune** for speeds/centring and **pulse** for short turns; **policy** sets corridor, dwell and normal stop/slow distances. Commands: README.

WHAT THE VGA IMAGE SHOWS

Upright camera image with red/blue contours and labels **R1/R2**, **B1-B6** or **BJ**. **CAL:W/B/Y** = waiting/calibrating/calibrated. **FRAME:GOOD/BAD/STALE** = image health. **STATE** = calibration interaction, not navigation.

SW6: RGB/HSV, colour, number or UNKNOWN, rejection code E. PAGE:1/2 alternates every 2 s across 16 slots, identified by yellow sample marks.

SW7: white 8 x 8 centre sample: RGB/HSV and the complete marker covering it. **SW8:** BYPASS:1 changes recognition, not camera view or safety. **SW9:** hides overlays.

READING THE SIX-DIGIT DISPLAY

page | mode | four data digits

d1 0520 = distance page, VISION_ONLY, **520 mm**. The second digit is the mode code above, not the bay number.

d	Millimetres. **FAr** = valid out-of-range, not measured 4 m. Four blank digits = invalid distance.
H	Last IR word, upper half in hexadecimal. Right pair **12** = POWER, **0F** = A (examples ED12 / F00F).
L	Last IR word, lower/address half in hexadecimal. H/L are blank until a complete valid IR word arrives.

KEY1: d - H - L - d. Left digit **n** = stable no-selection; **C** = stable multi-switch conflict, replacing the page letter. No ARM or driving-state page.

LEDs: 0 motion permitted; 1 IR stop latched; 3 docked; 8 target locked and fresh.

APPROACH / STOP Boot: 5 cm/s far, 4 cm/s near 500 mm; centre tolerance +/-30 px. Normal dock <=350 mm requires a new task to move again. Stationary **policy**: corridor45-70 px, dwell500-2000 ms, stop300-400 mm, slow500-1000 mm; boot45/1500/350/500. Fixed obstacle HOLD <=250 mm; release after three fresh >=300 mm readings. Invalid/stale range and IR A stop.

ADJUSTABLE SHORT TURN ALIGN kick3-8 cm/s for0-100 ms, then cruise3-5 cm/s. Total min/max20-300 ms includes kick; time steps10 ms. E14 boot: total200/200 ms, kick60 ms at6 cm/s, cruise4. A qualified gap may finish the original <=200 ms turn, without extension/re-kick. Faults, IR/J3 and a trusted crossing stop it. APPROACH loss stops; new-frame confirmation follows zero dwell.

SEARCH / RESET Boot scan3 cm/s,15.1 s is a historical plastic-floor time profile, not exact360 degrees. Up to three150 ms reverse steps share its budget. Search-speed changes invalidate qualification. KEY0 restores production values, colour calibration and zero-then-T131 initialization. Red2 must leave then re-enter. PC tune/pulse --defaults restore older generic values, not this boot profile. See README for exact ranges/steps.

ECHO-7 | R5 MEASURABLE TARGETS

Source **E7070014** | Prospective acceptance specification | 05 NOV 2026

Robot: OV7670 pixels -> DE1-SoC recognition/calibration -> target qualification and motion FSM -> UGV02 UART. SW0-SW5 choose a blue bay; IR A and J3 protect motion.

Claimed grade level: PENDING - no level asserted. Current three-trial scores are pending. Historical E4/E12 observations below do not certify this revision.

Required field	T1 - Marker accuracy	T2 - Vision latency	T3 - Colour calibration	T4 - Docking gap error
Quantity + unit	Correct blue-marker identities (%) = correct / 60 expected labels per trial.	Sampled capture-to-result latency (ms): unsigned completionUs - captureStartUs.	Confirmed calibration attempt to new profile commit (ms); conservative observed upper bound.	Absolute final gap error (mm): |ruler gap -350|, measured from J3 probe front to target plane.
Test conditions	Six official blue 1-6 signs, fixed and fully visible with >=10 px margin; camera 0.40-0.70 m away. Stable indoor light, no glare. VISION_ONLY; SW8=0, SW9=0; calibrated profile held fixed.	Same scene ranges as T1, nominal 25 fps and 320 x 240 RGB565. Use complete FPGA reports, including failed results; keep build/config fixed. No newly dropped/overflow frames during a scored trial.	Red1 plus its paired blue reference, both complete and still; 0.40-0.70 m camera distance, stable indoor light, nominal 25 fps. VISION_ONLY; same build; independent rearm before each attempt.	Already-visible blue 1-4, locked within U90-150; start gap 0.60-0.70 m. Flat plastic floor, vertical board spanning camera/J3 heights, unchanged supply/load. Fixed declared runtime settings for all 3 runs.
Three-trial scoring rule	Three tutor-chosen static poses. Per pose, score the first 10 distinct reports against a photographed label/region map: 6 labels x 10. Count bad/missing/ambiguous labels as wrong; no best-frame filtering. Use the LOWEST of three rates; require zero wrong accepted IDs.	Use T1's three sets of 10 distinct reports. Take each set's maximum board-time latency, then the LARGEST of those three. A missing report or new drop/overflow fails the trial. This sampled maximum does not bound unsampled frames or camera-to-wheel latency.	Three independently rearmed attempts. Log board-time trace from confirmed attempt through eight qualified samples and new-config commit. Use the LONGEST conservative interval. Failure, missing start/commit or >2 s is not a success; ACK alone does not count.	Reset to the same start marks for 3 approaches. Stop a trial after 25 s if no autonomous DOCKED. Measure each settled ruler gap; use the LARGEST absolute error. Any timeout/manual stop/no DOCKED fails; J3 alone cannot replace the ruler.
Threshold	>=95%; zero wrong accepted IDs; every bay appears in each trial.	<=150 ms, with all three sets complete and no new drop/overflow.	All 3 commits <=2000 ms, each with 8 valid sample frames.	3/3 autonomous docks; worst absolute error <=50 mm.
Goal	100% in all three trials.	<=100 ms under the same rule.	All 3 commits <=1000 ms.	3/3 docks; worst error <=20 mm.
Predicted value	100% intended on the declared static sheet. Design expectation, not a proven physical success rate.	Provisional 100 ms budget: 40 ms capture + 60 ms processing/commit allowance. Not a proven worst-case bound.	Ideal 360 ms: (8 sample frames + 1 commit boundary) x 40 ms. Excludes pre-confirmation detection and rejected frames.	**PENDING** braking/ranging error model. A logical 350 mm stop threshold is not a physical gap prediction.
Measured value	E14: **PENDING**. E4 history: all six IDs present in 12/12 snapshots in each of three sets; this is not spatially truth-scored T1 accuracy.	E14: **PENDING**. E12 history: 38.818-57.066 ms across 90 sampled reports in one moving-target task; not the required three static sets.	E14: **PENDING**. E12 preflight showed calibrated=true and 8 sample frames; three independently timed attempts are absent.	E14: **PENDING**. Latest E12 outside-view task ended INCOMPLETE at 60.007 s, not DOCKED; no valid T4 error. Earlier partial runs cannot be joined.
Claimed grade level	**PENDING**	**PENDING**	**PENDING**	**PENDING**

Common rule: The tutor chooses within each range. Record all three trials; never select the best. T1/T2 may share captures. Use first distinct reports, retain failed/missing results and reject duplicate frame IDs. Plan about 90 s per target; T4 has a 25 s/run cap. Demonstrate IR safety before driving.

Prediction / scope: Estimates are prospective, not historical predictions. T2 excludes scene-to-UART and mechanical latency. T4 covers already-visible targets only; a complete outside-view search-to-dock demonstration remains separate and unpassed. Do not join partial runs or treat command speeds as measured wheel speeds.

Contribution / R9: T1/T2 cover the projective/anchor reader and perception pipeline; T3 covers real-pixel calibration; T4 covers target qualification and motion decisions. Repeat T1 with SW8=1 under matched conditions. If bypass also meets threshold, this target does not establish that stage's necessity.

Sources: Major_Project_Brief_2026.pdf: pp 3-4 R5/2.3 (target rules), pp 4/6 2.4/4 (level and evidence), p 7 submission. All numeric thresholds/goals are project choices: T1-T3 retained from render_member_b_datasheet.py; T4 newly declares 350 mm +/-50 mm threshold /+/-20 mm goal. Detailed provenance: docs/R5_TARGETS.md; editable specification: tools/data/echo7_r5_targets.json.